**Joseph Meister** a nine year old boy, severely bitten by a rabid dog and in grave risk of developing rabies, was given a course of **13 inoculations** of the infected cord **vaccine** by Pasteur. The boy **survived**. This dramatic event was a **mile stone** in the development of medicine.

## Morphology

The rabies virus is **bullet shaped**, with one end rounded or conical and the other planar or concave. The lipoprotein envelope, carries **knob like spikes**, composed of **glycoprotein G** responsible for pathogenesis, virulence and immunity beneath the envelope is the **matrix (M) protein layer** which may be invaginated at the planar end. The membrane may project outwards forming a **bleb**. The genome is unsegmented linear RNA (Figure 10.6).

The rabies virus isolated from natural human or animal infection is termed **'the street virus'**. Rabies has been recognized from very ancient times as a disease transmitted to humans and animals by the **bite of 'mad dogs'**. The name rabies comes from the **Latin** word rabidus, meaning **'mad'**, derived from the **Sanskrit** root rabhas, for **frenzy**.

## Pathogenesis

Human infection is usually caused by the **bite of rabid dogs** or other animals. The virus present in the saliva of the animal is **deposited** in the wound (Figure 10.7). **Rarely**, infection can also occur following non-bite exposures such as licks or aerosols.

The virus appears to **multiply** in the muscles, connective tissue or nerves at

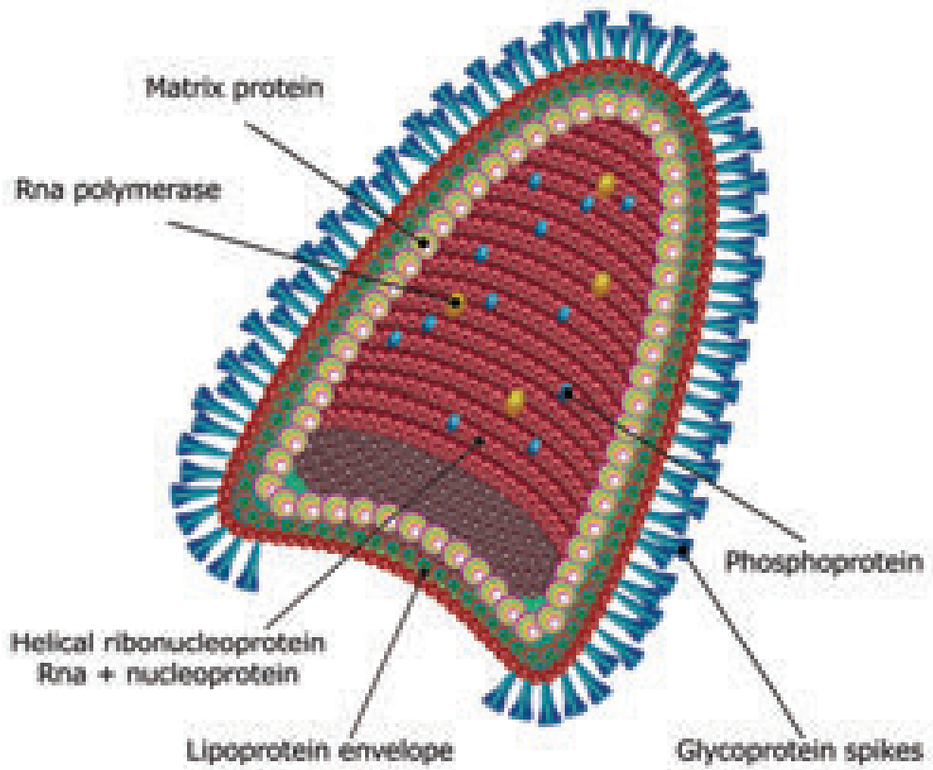



**Figure 10.6:** Structure of Rabies Virus



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